Lecture 12 Compendium: Practical Drone Operations for Surveillance

Module 4: Practical Skills & Applied AI

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Compendium for Advanced Study

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1 Introduction: The Mission Commander Mindset

In theoretical modules, we treated the drone as an abstract sensor platform. In this practical module, we confront the reality of **Field Operations**.

As a PhD student conducting field research, you are not just a "Drone Pilot"; you are a **Mission Commander**. You are responsible for:

1. **Safety:** Ensuring no harm to people or property.
2. **Legality:** Adhering to Tanzania Civil Aviation Authority (TCAA) regulations.
3. **Data Integrity:** Ensuring the images captured are scientifically usable (Sharp, Exposed, Geotagged).

The Cost of Failure

If you fly a mission and the images are blurry, you have wasted a day of battery charging, travel costs, and field labor. Worse, you may miss a critical phenological stage (e.g., flowering) that will not return for a year. **Validation in the field is non-negotiable.**

2 Mission Mathematics: Planning the Flight

Before leaving the lab, you must mathematically calculate your flight parameters.

2.1 Ground Sampling Distance (GSD)

GSD is the distance between the centers of two consecutive pixels measured on the ground. It defines the **Spatial Resolution** of your data.

$$GSD \text{ (cm/px)} = \frac{S_w \times H \times 100}{F_r \times I_w} \tag{1}$$

Where:

- S_w: Sensor Width (mm)
- H: Flight Altitude (m)
- F_r: Focal Length (mm)
- I_w: Image Width (pixels)

Example Calculation:

- **Drone:** DJI Phantom 4 Pro ($S_w = 13.2$mm, $F_r = 8.8$mm, $I_w = 5472$px).
- **Altitude:** 100m.

$$GSD = \frac{13.2 \times 100 \times 100}{8.8 \times 5472} \approx 2.74 \text{ cm/px}$$

2.2 Photogrammetric Overlap

To create a 3D model (Orthomosaic) using Structure from Motion (SfM), the software must identify common "tie points" between images.

- **Front Overlap:** Along the flight path. Typically **75-80%**.
- **Side Overlap:** Between parallel flight lines. Typically **60-70%**.

Operational Trade-off: Higher overlap = Better model quality, but significantly longer flight time and more data processing.

3 Python Tool: The Mission Planner

We will implement a Python class to calculate mission parameters automatically.

```python
class MissionPlanner:
    def __init__(self, sensor_w_mm, focal_len_mm, img_w_px, img_h_px):
        self.Sw = sensor_w_mm
        self.Fr = focal_len_mm
        self.Iw = img_w_px
        self.Ih = img_h_px

    def get_gsd(self, altitude_m):
        """Returns GSD in cm/pixel"""
        return (self.Sw * altitude_m * 100) / (self.Fr * self.Iw)

    def get_altitude(self, target_gsd_cm):
        """Returns required altitude in meters for a target GSD"""
        return (target_gsd_cm * self.Fr * self.Iw) / (self.Sw * 100)

    def get_coverage(self, altitude_m, front_overlap=0.80, side_overlap=0.70):
        """
        Calculates the physical footprint of one image and
        the trigger distance/line spacing.
        """
        # Footprint on Ground (meters)
        footprint_w = (self.Sw * altitude_m) / self.Fr
        footprint_h = (self.Sw * (self.Ih / self.Iw) * altitude_m) / self.Fr

        # Distances to travel before taking next photo
        trigger_dist = footprint_h * (1 - front_overlap)
        line_spacing = footprint_w * (1 - side_overlap)

        return {
            "Footprint": (footprint_w, footprint_h),
            "Trigger_Interval": trigger_dist,
            "Line_Spacing": line_spacing
        }

# --- EXAMPLE USAGE ---
# DJI Mavic 2 Pro (Hasselblad)
drone = MissionPlanner(13.2, 10.26, 5472, 3648)

# Requirement: We need < 1 cm/pixel to count individual leaves
required_alt = drone.get_altitude(1.0)
print(f"Required Altitude: {required_alt:.1f} meters")

# Plan grid
plan = drone.get_coverage(required_alt)
print(f"Distance between flight lines: {plan['Line_Spacing']:.1f} meters")
```

Listing 1: Drone Mission Planner Implementation

4 Operational Workflows: The Checklists

Aviation safety is built on checklists. Memorization kills.

4.1 Phase 1: Pre-Site Planning (Office)

1. **Site Survey:** Check Google Earth. Identify hazards (Powerlines, tall trees).
2. **Airspace Check:** Are we near an airport? (No Fly Zone).

3. **Weather:** Check wind speed (< 10 m/s) and KP Index (Solar activity). High KP (> 4) affects GPS accuracy.
4. **Hardware:** Charge all batteries. Format SD cards. Update Firmware.

4.2 Phase 2: On-Site Pre-Flight

1. **Visual Inspection:** Check propellers for micro-cracks. Check motors for grit.
2. **Calibration:** Calibrate the compass (away from metal/cars).
3. **Home Point:** Verify the "Home Point" is recorded correctly on the map.
4. **RTH Altitude:** Set "Return to Home" altitude higher than the tallest obstacle in the area (e.g., 50m).

4.3 Phase 3: In-Flight Monitoring

1. **VLOS (Visual Line of Sight):** Keep eyes on the drone, not just the screen.
2. **Battery Monitoring:** Plan to land with 20% battery remaining. Do not push to 0%.
3. **Link Quality:** If the RC signal drops, ensure the antenna flat side faces the drone.

5 Troubleshooting Field Failures

5.1 The "Toilet Bowl Effect" (TBE)

Symptom: The drone drifts in circles, getting wider and wider, unable to hold a hover position. **Cause:** Compass Interference. The drone's internal compass disagrees with the GPS vector. **Solution:** 1. **IMMEDIATE:** Switch flight mode from "P-GPS" to "Atti" (Attitude Mode). This disables GPS/Compass reliance and relies only on the Gyroscope. The drone will drift with the wind, but it will stop circling. 2. **Land Manually.** 3. **Recalibrate Compass** in a new location away from magnetic interference (e.g., rebar in concrete, speakers, cars).

5.2 Motion Blur

Symptom: Images look smeared. **Cause:** Shutter speed is too slow for the drone's ground speed. **Solution:** 1. Increase Shutter Speed (e.g., 1/1000s or faster). 2. Slow down the drone flight speed. 3. Increase ISO (but watch for noise).

6 Data Validation: The Histogram

How do you know if your images are good without downloading them to a laptop? **Check the Histogram.**

- **Underexposed:** The graph is bunched to the left (Blacks). Shadows contain no data.
- **Overexposed:** The graph is bunched to the right (Whites). Highlights are "blown out."
- **Correct Exposure:** The graph is a "bell curve" in the middle, or spread evenly.

PhD Rule: Always fly with "Zebra Stripes" (Overexposure Warning) turned ON. If you see stripes on the crop canopy, your data is invalid.

7 Regulatory Compliance (TCAA Context)

In Tanzania, drone operations are regulated by the **Civil Aviation (Remotely Piloted Aircraft Systems) Regulations**.

1. **Registration:** All drones weighing > 250g must be registered with TCAA.
2. **Licensing:** Commercial/Research operations often require a Remote Pilot License (RPL).
3. **Permits:** A specific permit is required for each commercial operation.
4. **Insurance:** Third-party liability insurance is mandatory.
5. **Privacy:** It is illegal to film private property without consent.

BVLOS (Beyond Visual Line of Sight): Flying where you cannot see the drone. This is strictly prohibited without a special waiver requiring Detect-and-Avoid radar systems.

8 Review Questions & Field Assignment

1. **Calculation:** You need to map a 50-hectare field with a GSD of 2 cm/px. Your drone flies at 10 m/s. The flight lines are spaced 20m apart.
   - Calculate the total flight distance.
   - If the battery lasts 20 minutes, how many batteries do you need?
2. **Scenario:** You are flying a mission and a hawk starts circling your drone aggressively. What is the correct physiological and operational response?
3. **Regulation:** You want to map a farm that is 3km away from the Julius Nyerere International Airport (JNIA). Is this legal? What specific TCAA authorization would you need?

References

1. TCAA. (2018). *Civil Aviation (Remotely Piloted Aircraft Systems) Regulations*. Government Printer.
2. Pix4D. (2020). *The Photogrammetry Guide: Flight Planning*.
3. Wolf, P. R., & Dewitt, B. A. (2014). *Elements of Photogrammetry with Applications in GIS*. McGraw-Hill Education.